

From Composition to Architecture-Aware Modeling

Stage 2A

Frac

$$h_{poly} = f_A h_A + f_B h_B$$

Baseline:
Weighted average
of monomer reps

+arch info



Stage 2D0

Global Architecture

$$h_{mix} + \alpha_{arch} \cdot e_{arch}$$

Architecture offset:
Learned embedding
per arch type

+chemistry



Stage 2D1

Chemistry-Conditioned

$$h_{mix} + \alpha_{arch} \cdot r_{arch}(h_A, h_B, \dots)$$

Architecture + chemistry:
Interaction MLP conditions
on monomer identity